- Mid-term pages:2-55
- Final exam pages: mid&(56-88)



NUMERICAL

The final exam material

Keep calm and iterate - sometimes you won't get it right on the first try, but that's what numerical methods (and life) are all about.

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Numerical method

26 Feb

Numerical is a method are techniques to solve math problems



Math formulation عن طريق الرسم
of the Problems

- The solution must be accurate and very close to exact solution (true solution)

Subject areas covered in this course

1. Errors

True = 2.134 القيمة الحقيقية

Approx = 2.213 القيمة التقريبية

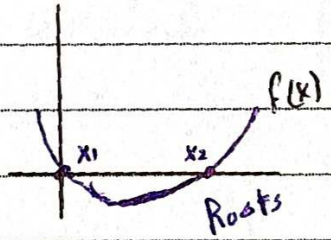
2. Roots

$X_1, X_2, \ldots$

أصفار المعادلة

$f(x_1) = 0$

$f(x_2) = 0$



26 Feb

3. Solution of simultaneous equation.

المعادلات الخطية المتزامنة هي مجموعة من المعادلات التي تحتوي على نفس المتغيرات ويجب حلها معًا لإيجاد القيم التي تحقق جميع المعادلات في نفس الوقت.

$$a_{11}x_1 + a_{12}x_2 = b_1$$

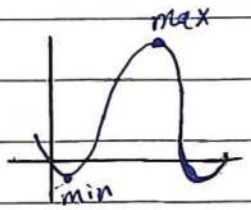
$$a_{21}x_1 + a_{22}x_2 = b_2$$

Solution: ① Graphical method ② Metrices method

4. Differential Equations

5. Optimization

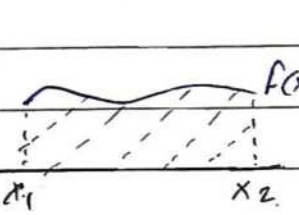
↳ max ↳ Min



6. Integration

المساحة تحت المنحنى

Area under the curve



$$\int_{x_1}^{x_2} f(x)\,dx$$

7. Curve fitting

$y = a_0 + a_1 x$ $y = a_0 + a_1 x + a_2 x^2$

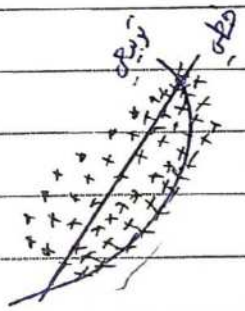
$R^2_{linear} = 0.862$ $R^2_{quadratic} = 0.9731$

0.9731 uncertainty is removal

↳ accuracy = 97.31 %

$\varepsilon = 2.69\%$

الأكثر دقة ← الأقل



R: Regression coefficient

معامل الارتباط

accuracy = 86.2 %

$\varepsilon = 13.8\%$

error ↵

يتبع

26 Feb

Curve Fitting

R → تدل على الربط بين القيم
كلما كبرت R
أصبح الترابط أكبر

Math modelling

Dependent Variable = f(dep. & indep. variable, Parameters, Forcing Function)

$F = ma$

F = Forcing function

m = Parameters (material Property)

a = dependent variable f(t,v)

* أي اقتران بالنسبة للزمن هو (dynamic)

Significant figures

Room	1m = 100cm 1cm = 10 mm

10.30 m
10.125 m
10.2 m
10.1150 m

→ أيها أنسب لقياس غرفة ؟

* نختار عدد الخانات التي تتسع لـ (m, cm, mm) أي أن الرقم المناسب هو (10.125)

* لو كان المثال عن مسافة مطلوبة بالسيارة ← نهمل خانات الـ cm, mm

Errors

3 Mar

True value = Approximate + Error

Error = True Value - Approximate

$$\varepsilon_t\% = \text{Error}\% = \left|\frac{\text{Error}}{\text{True value}}\right| \times 100\%$$

$\varepsilon_t\%$: Absolute Error %

$$\varepsilon_{t1} = \frac{T - T_1}{T} \times 100\% \qquad \varepsilon_{t2} = \frac{T - T_2}{T} \times 100\%$$

$$\varepsilon_{t3} = \frac{T - T_3}{T} \times 100\% \qquad \varepsilon_{t\infty} = \frac{T - T_\infty}{T} \times 100\%$$

$$\varepsilon_a\% = \left|\frac{\text{New value} - \text{Old value}}{\text{New value}}\right| \times 100\%$$

$\varepsilon_a\%$ = Relative Error %

$$\varepsilon_{a1} = \left|\frac{\text{New} - \text{old}}{\text{New}}\right| \times 100\% \rightarrow \text{old} = 0 \rightarrow$$ لا يوجد قيمة سابقة

$$\varepsilon_{a1} = 100\% \qquad \varepsilon_{a2} = \left|\frac{T_2 - T_1}{T_2}\right| \times 100\%$$

$$\varepsilon_{a3} = \left|\frac{T_3 - T_2}{T_3}\right| \times 100\%$$

Errors 3 Mar

$$\varepsilon_s\% = (0.5 \times 10^{2-n})\%$$

$\varepsilon_s\%$: Prespecified error %

n: Number of significant Numbers in the calculated error

ε_a &/or $\varepsilon_t < \varepsilon_s$

Example: Compute $e^{0.5}$ using the following function:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!}, \quad n = 3$$

Solution:

True Value: $e^{0.5} = 1.648721$ ← عن طريق الآلة الحاسبة

No. of term	$e^{0.5}$	$\varepsilon_t\%$	$\varepsilon_a\%$
1	1	39.3%	100%
2	1.5	9.02%	33.33%
3	1.625	1.44%	7.69%
4	1.64583	0.175%	1.27%
5	1.64843	0.0172%	0.158%
6	1.64869	0.00142%	0.0158%

1.64869 ← عن طريق التعويض بالقوانين

$$R_n = (1.648721 - 1.64869)$$

True value — Calculated

3 Mar

No. of term: عدد الحدود في الإكسبان (e^x)

عند الـ (third term) نعوض $e^{(0.5)}$ في أول ثلاثة حدود

$$1 + e^{0.5} + \frac{(e^{0.5})^2}{2!} = 1.625$$

$n = 3$

$\varepsilon_s\% = 0.5 \times 10^{2-3}\ \%$

$\varepsilon_s\% = 0.05\ \%$

نتوقف عند الـ term الذي يكون فيه $\varepsilon_t / \varepsilon_a$ اقل من ε_s

← ذلك في حل طلبي لسان $\varepsilon_t / \varepsilon_a$

← لو كان المطلوب فقط ε_t ← نتوقف عندما $\varepsilon_s > \varepsilon_t$

$$\varepsilon_{t1} = \frac{1.648721 - 1}{1.648721} \times 100\% \rightarrow \varepsilon_{t1} = 39.3\%$$

$$\varepsilon_{t2} = \frac{1.648721 - 1.5}{1.648721} \times 100\% \rightarrow \varepsilon_{t2} = 9.02\%$$

$\varepsilon_{a1} = \frac{1 - 0}{1} \times 100\%$	$\varepsilon_{a2} = \frac{1.5 - 1}{1.5} \times 100\%$
$\varepsilon_{a1} = 100\%$	$\varepsilon_{a2} = 33.33\%$

3 Mar

Truncition Error(R_n) & Tylor Series

R_n: Remainder

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + R_n$$

(True value) هذا توقفنا عند (6) و تبقى خطأ بسيط حتى نصل لـ

Truncition هذا ما نسميه بالـ

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \dots + R_n$$

$$h = x_{i+1} - x_i$$

الفرق (1)

$h = 1$ one interval

h: interval or step size

In the Previous example: find R_n

R_n = 1.648721 − 1.64869

True value — calculated (6th term)

3 Mar

Example: use zero to 4th order of Tylor series to find the value of the function:

$f(x) = -0.1x^4 - 0.15x^3 - 0.25x + 1.2 - 0.5x^2$

at $x_{i+1} = 1$, The value of $x_i = 0$

Solution:

$h = x_{i+1} - x_i$

$h = 1 - 0$

$h = 1$

True value: $f(x_{i+1})$

$= f(1)$

$f(1) = -0.1 - 0.15 - 0.5 - 0 - 0.25 + 1.2$

$f(1) = 0.2$

[1] Zero order

→ $f(x_{i+1}) = f(x_i)$ الحد الأول من معادلة Tylor

$f(1) = f(0)$

$f(1)$ $0 - 0 - 0 - 0 + 1.2$

$f(1) = 1.2$

Error → True value - approximation

Error $= 0.2 - 1.2 = -1$

$$\varepsilon\% = \left|\frac{0.2 - 1.2}{0.2}\right| \times 100\% = 500\%$$

3 Mar

[2] 1st order

$\rightarrow f(x_{i+1}) = f(x_i) + f'(x_i)h$

$f(1) = f(0) + f'(0)h$

$f(1) = 1.2 + (-0.25)(1)$

$f(1) = 0.95$

$f'(x_i) = -0.4x^3 - 0.45x^2 - x - 0.25$

$f''(x_i) = -1.2x^2 - 0.9x - 1$

$\text{Error} = 0.2 - 0.95 = -0.75$

$\varepsilon\% = \left|\frac{0.2 - 0.95}{0.2}\right| \times 100\%$

$\varepsilon\% = 375\%$

$f'''(x_i) = -2.4x - 0.9$

$f^{(4)}(x_i) = -2.4$

[3] 2nd order

$\rightarrow f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!}$

$f(1) = f(0) + f'(0)(1) + \frac{f''(0)(1)^2}{2}$

$f(1) = 0.45$

$\text{Error} = 0.2 - 0.45 = -0.25$

$\varepsilon\% = 125\%$

[4] 3rd order

$$\rightarrow f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!}$$

$$f(1) = f(0) + f'(0)h + \frac{f''(0)h^2}{2!} + \frac{f'''(0)h^3}{3!}$$

$f(1) = 0.3$

$f'''(x_i) = -2.4x - 0.9$

$f'''(x_i) = -2.4$

Error $= 0.2 - 0.3 = 0.1$

$$\varepsilon\% = \left|\frac{0.2 - 0.3}{0.2}\right| \times 100\% \qquad \varepsilon\% = 50\ \%$$

[5] 4th order

$$\rightarrow f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \frac{f''''(x_i)h^4}{4!}$$

$$f(1) = f(0) + f'(0)h + \frac{f''(0)h^2}{2!} + \frac{f'''(0)h^3}{3!} + \frac{f''''(0)h^4}{4!}$$

$f(1) = 0.29 \quad 0.2$

Error $= 0.2 - 0.29 = -0.09$ Zero

$$\varepsilon\% = \left|\frac{0.2 - 0.29}{0.2}\right| \times 100\ \% \qquad \varepsilon\% = 45\%$$

Zero

Root of equations

5 Mar

Roots of $f(x)=0$ at a given x-value

$f(x) = (x+1)(x-2)$, find the roots

$f(x) = 0$

$x = -1, \quad x = 2$

Method of Solution:

① **Analytical Method**

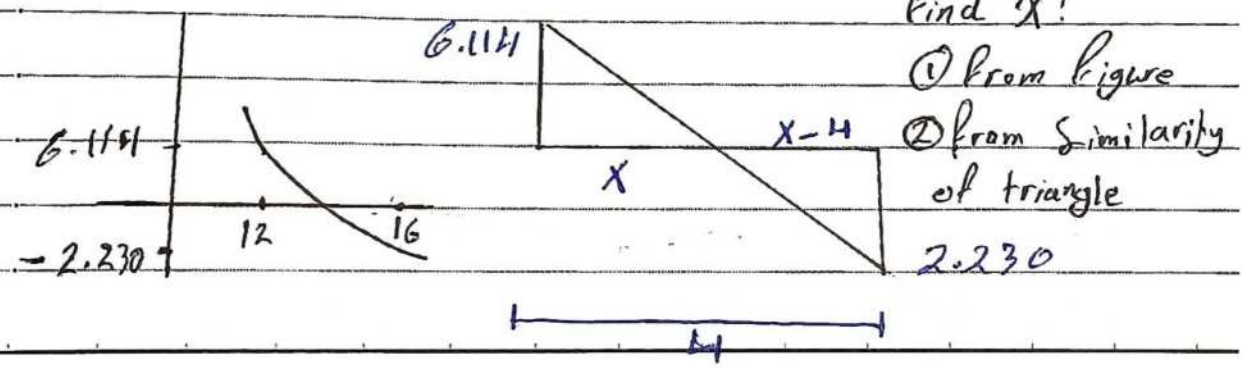
→ $f(x) = ax^2 + bx + c$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

② **Graphical** حصر الصفر بين أول قيمة موجبة وأول قيمة سالبة

$$f(x) = \frac{668.06}{x}\left(1 - e^{-0.146843x}\right) - 40$$

x	f(x)	
4	34.19	
8	17.712	
12	6.114	] الصفر بينهم
16	-2.230	]
20	-8.368	



find x:

① From figure

② From Similarity of triangle

5 Mar

③ Numerical Methods

- Bracketing Method (closed)
  - Bisection Method
  - False Position
- Open Method
  - Simple fixed Method
  - Newton Raphson Method
  - Secant Method

Bisection Method

Algorithm: $X_{Lower} < X_{root} < X_{upper}$

$$X_r = \frac{X_L + X_u}{2}$$

IF: $f(X_L) \cdot f(X_r)$

< 0, set $X_u = X_r$

> 0, set $X_L = X_r$

$= 0$, set X_r is the root

5 Mar

Example:

$f(x) = \sin x + x^2 + x - 10$

Find the root between 2 & 4

Solution:

$x_L = 2, \quad x_u = 4, \quad x_r = \frac{x_L + x_u}{2}$

$x_{r1} = 3$

$f(2) = \sin(2) + (2)^2 + 2 - 10 = -3.081$

$f(3) = \sin(3) + (3)^2 + 3 - 10 = 2.141$

$f(x_L) \cdot f(x_r) < 0$ ✱

→ set $x_u = x_{r1} = 3$

و $x_L = 2$ كما هي

$x_{r2} = \frac{2+3}{2}, \quad x_{r2} = 2.5$

$f(2) = -3.081$

$f(2.5) = -1.65$

$f(x_L) \cdot f(x_r) > 0$ ✱

→ set $x_L = x_r = 2.5$

$x_u = 3$

$\rightarrow x_{r3} = 2.75$

$x_{r1} - x_{r2} \quad \varepsilon_a\% = \left|\frac{2.5 - 3}{2.5}\right| \times 100\% \quad \varepsilon_a\% = 20\%$

$x_{r2} - x_{r3} \quad \varepsilon_a\% = \left|\frac{2.75 - 2.5}{2.75}\right| \times 100\% \quad \varepsilon_a\% = 9.09\%$

and so on → لحد ما أقترب من الصفر أو على حسب المطلوب

5 Mar

Newton Raphson Method

Algorithm: $x_a < x_0 < x_b$

$$x_0 = \frac{x_a + x_b}{2}$$

$$x_1 = x_0 - \left(\frac{f(x_0)}{f'(x_0)}\right), \quad x_2 = x_1 - \left(\frac{f(x_1)}{f'(x_1)}\right), \text{ ... and so on}$$

Example: find the root of the function $f(x) = -4 + \cos x + 2$ between 0 & 1.0

Solution:

$f'(x) = -4 - \sin x$

$x_0 = \frac{0+1}{2} = 0.5$

$f(0.5) = 0.8776 \quad f'(0.5) = -4.4794$

$x_1 = 0.5 - \frac{f(x_0)}{f'(x_0)} = 0.6959$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad x_2 = 0.6924$

$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}, \quad x_3 = 0.6924$ is a root ← متساويان

$$\varepsilon_{a_1} = \frac{0.6959 - 0.5}{0.6959} \times 100\% = 28.1\%$$

$\varepsilon_{a_3} = 0\%$

$$\varepsilon_{a_2} = \frac{0.6924 - 0.6959}{0.6924} \times 100\% = 0.5\%$$

Lecture(4) - Roots

10 March

Newton's Raphson method

$$X_{i+1} = X_i - \frac{f(X_i)}{f'(X_i)}$$

N.R method converges if $f(x) = 0$ & $|f(x)\,f''(x)| < (f'(x))^2$

Roots of Polynomials

→ Division of functions:

1. Long Division القسمة الطويلة
2. Synthetic Division القسمة التركيبية

Synthetic Division

$$P(x)/q(x) = Q(x) + R/q(x)$$

$P(x)$: Dividend المقسوم

$q(x)$: Divisor المقسوم عليه

$Q(x)$: Quotient الناتج

R: Remainder الباقي

1o March

$P(x)/q(x) = Q(x) + R/q(x)$

$\rightarrow \frac{10}{3} = 3 + \frac{1}{3}$

$\rightarrow \frac{13}{4} = 3 + \frac{1}{4}$

Synthetic Division

Ex. Divid $\overset{P(x)}{4x^3 - 2x^2 + 3x + 1}$ by $\overset{q(x)}{x-2}$

Solution: $x-2=0 \rightarrow (x=2)$

x^3	x^2	x	constant
4	-2	3	1
②	8	12	30
4	6	15	31 → الباقي

$\frac{P(x)}{q(x)} = ax^2 + bx + c + R$

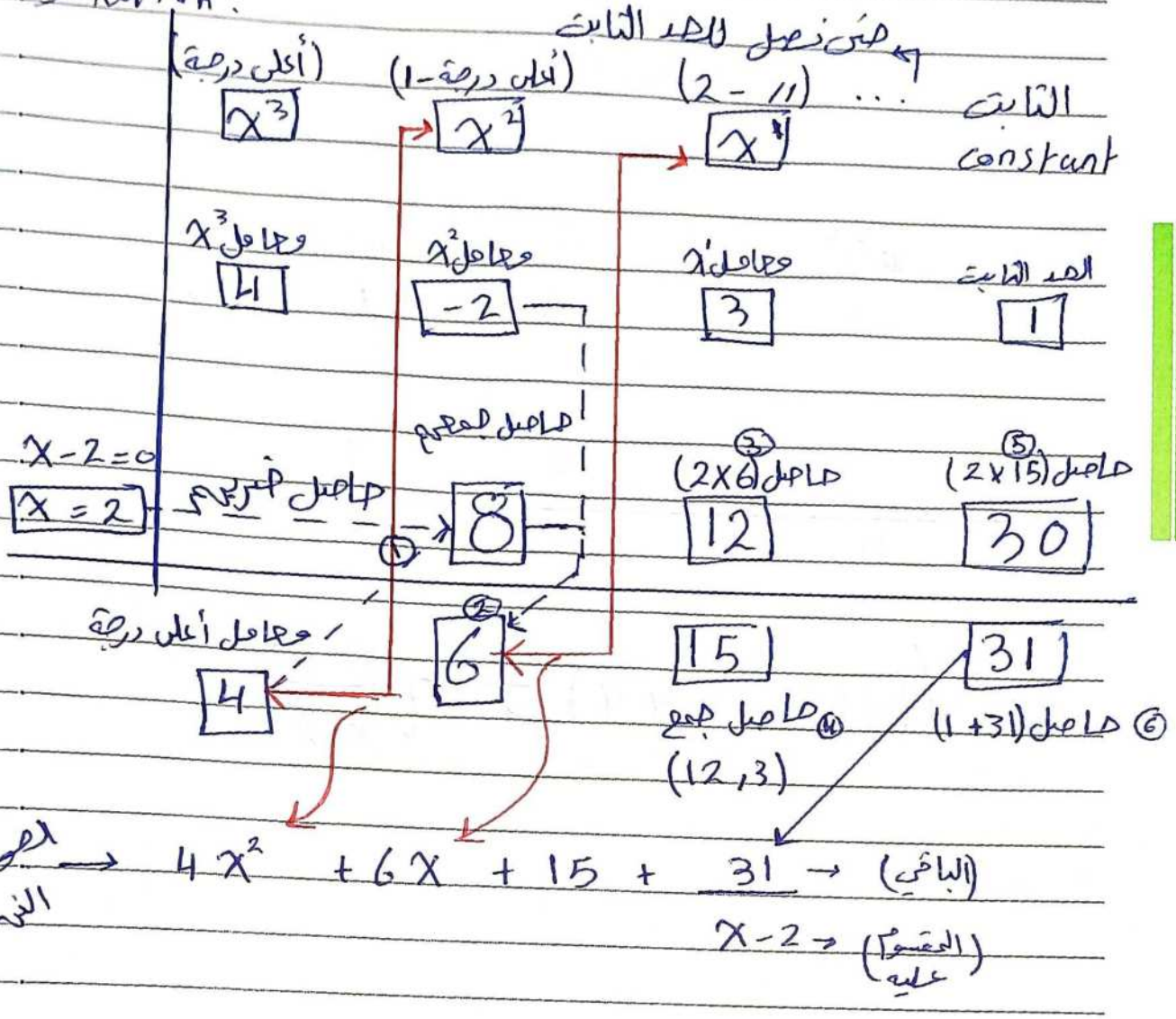
$\rightarrow \frac{4x^3 - 2x^2 + 3x + 1}{x-2} = 4x^2 + 6x + 15 + \frac{31}{x-2}$

10 March

شرح المثال السابق :-

Divid $4x^3 - 2x^2 + 3x + 1$
by $x - 2$

Solution:



1o March

Ex. Divid $x^4 + 3x^2 + 1$ by $x^2 - 2x + 3$ (p, q)

Solution:

	x^4	x^3	x^2	x	constant
	1	0	3	0	1
$(-p)$ 2	=	2	4	8	
$(-q)$ -3	=	-3	-6	-12	
	1	2	4	2	-11

$\rightarrow x^2 + 2x + 4$ $\qquad R = 2x - 11$

$$\frac{P(x)}{q(x)} = x^2 + 2x + 4 + \frac{2x - 11}{x^2 - 2x + 3}$$

* في حال القسم على $2x^2 - 2x + 3$

$\hookrightarrow 2(x^2 - x + \frac{3}{2})$

$\hookrightarrow$ ** بضرب البرهنة بـ $\frac{1}{2}$

10 March

Ex. Divid $3x^4 - 2x^2 + 4$ by $2x+2$

Solution:

$2x+2$

$2(x+1)$

$x = -1$ العكس

x^4	x^3	x^2	x	constant
3	0	-2	0	4
-1	-3	3	-1	1
3	-3	1	-1	5

$$\rightarrow \frac{3x^4 - 2x^2 + 4}{2x+2} = \left(3x^3 - 3x^2 + x - 1\right)\frac{1}{2} + \frac{5}{2x+2}$$

Lecture(5)

Roots of Polynomials Pns

12 March

Bairstow's Method

Method algorithm:-

1. Arrange the Pn in the form

$$P(x) = a_1 x^n + a_2 x^{n-1} + \cdots + a_0$$

2. select the quadratic factor equation

$$x^2 + px + q$$

3. Conduct the 1st Synthetic Division

(a-values)	a_1	a_2	a_3	$\cdots$	a_0
$-p$	–	$-pb_n$	$-pb_{n-1}$		☐ → الرقم الأخير يؤثر فقط على الباقي
$-q$	–	–	$-qb_n$		
(b-values)	b_n	b_{n-1}	b_{n-2}		

12 March

4. conduct the second Synthetic Division (S.D)

b-values	b_n	b_{n-1}	b_{n-2}	....	b_0
$-p$	—	$-pC_n$	$-pC_{n-1}$		
$-q$	—	—	$-qC_n$		
c-values	C_n	C_{n-1}	C_{n-2}	...	

5. Write the following two equations & Solve them

$$C_2 S_p + C_3 S_q = b_1, \quad (C_1 - b_1) S_p + C_2 S_q = b_0$$

S_p or S_q : The Partial derivative of the remainder S with respect to p or q. It measures how S changes when p or q changes. المشتقة الجزئية للباقي S بالنسبة لـ (p أو q) وتقيس تغيّر S عند تغيير (p أو q)

6. Write the quadratic equation

$$x^2 + (p + S_p)x + (q + S_q) = 0$$
$$\hookrightarrow x^2 + p_2 x + q_2$$

$$x = \frac{-p_2 \pm \sqrt{(p_2)^2 - 4q_2}}{2}$$ القانون العام

12 March

Example: Use Bairstow's method to find the complex roots of $f(x) = 4 + x^2 - x + x^3$

→ use quadratic equation : $x^2 - 3x - 1$

Solution:

$f(x) = x^3 + x^2 - x + 4$ $\quad (x^2 - 3x - 1)$

$p = -3$

$q = -1$

	x^3	x^2	x	constant
	1	1	-1	4
3		3	12	36
1			1	4
	1	4	12	44
	b_3	b_2	b_1	b_0

	x^2	x	constant
	1	4	12
3		3	21
1			1
	1	7	34
	c_3	c_2	c_1

في حال كانت المعادلة $2x^2 - 3x - 1$

↳ بقسم الـ b-Values على 2

① $(7)\delta p + (1)\delta q = 12$, $\quad (34-12)\delta p + (7)\delta q = 44$

② $(22)\delta p + (7)\delta q = 44$

نضرب معادلة ① بـ (-7) وتضعها للمعادلة ②

$\delta p = 1.482$

$\delta q = 1.630$

$x^2 + (p + \delta p)x + (q + \delta q)$

→ $x^2 - 1.518x + 0.630 = 0$

$x = \underbrace{0.759}_{\alpha \text{ alpha}} \pm \underbrace{0.232}_{\beta \text{ Beta}} i$

Linear Algebric Equation

17 March

For this system of Eqns:-

$a_{11} x_1 + a_{12} x_2 = b_1$

$a_{21} x_1 + a_{22} x_2 = b_2$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

[A] [X] [B]

2*2 2*1 2*1

(يجب أن تتساوى هذه الأعداد) → Matrix size → size of the result

m * n

m: number of rows

n: number of columns

A * B = C

4*3 3*2 4*2 (3)(1) + (1)(-1) + (1)(-3) = -10

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} = \begin{bmatrix} -10 \\ -18 \end{bmatrix}$$

2*3 3*1 2*1

(2)(1) + (2)(-1) + (6)(-3)

= -18

17 March

$$A = [2 \quad 1 \quad -2 \quad 3]$$

Row matrix

$$A = \begin{bmatrix} 5 \\ 3 \\ -2 \\ -4 \end{bmatrix}$$

Column matrix

→ Vector matrices

كاعود واحد or صف واحد

Diagonal matrix

عدد الصفوف = عدد الأعمدة

المصفوفة القطرية

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Identity matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Diagonal (1)

$$[A]^{-1}[A] = 1$$

$[A]^T$: The transpose of $[A]$

$$A = \begin{bmatrix} 2 & 1 & -4 \\ 3 & 3 & 5 \\ 1 & 2 & 3 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 3 & 2 \\ -4 & 5 & 3 \end{bmatrix}$$

17 March

Upper triangular matrix

$$\begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

lower triangular matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 6 & 3 & 3 \end{bmatrix}$$

Determinants

For 2 x 2 matrix A: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$|A| = \text{determinant.}(A) = a_{11}a_{22} - a_{12}a_{21}$

For 3 x 3 matrix A: $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$\text{det.}(A) = +a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

IF det. A = 0, A is a Singular matrix

17 March

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 2 & -3 \\ 0 & 0 & 5 \end{bmatrix}$$

Det. of upper & lower = Multiplicate of diagonal

$1 * 2 * 5$

Matrix Inverse

1) 2X2 matrix $A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$

$$[A]^{-1} = \frac{1}{\det A} \begin{vmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{vmatrix}$$

to check : $[A]^{-1}[A] = I$

Example : Find the Inverse of $\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix}$

det. $= 2 * 3 - 1 * 2 = 4$

$$A^{-1} = \frac{1}{4} \begin{vmatrix} 3 & -1 \\ -2 & 2 \end{vmatrix}$$

$$\text{check} = A^{-1}A = \frac{1}{4} \begin{vmatrix} 3 & -1 \\ -2 & 2 \end{vmatrix} * \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix}$$

$$= \frac{1}{4} \begin{vmatrix} 4 & 0 \\ 0 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

17 March

2) Inverse of 3*3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Solution:

① find det of A

② find minor cofactors a_{ij}

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

⋮

$$M_{33} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

③ find major cofactors

Cofactors $m_{ij} = (-1)^{i+j} M_{ij}$ (minor)

Assume:

$$\begin{bmatrix} 13 & -16 & 17 \\ 21 & 6 & 5 \\ 9 & 8 & 6 \end{bmatrix} = \begin{bmatrix} 13 & 16 & // \\ // & // & // \\ // & // & // \end{bmatrix}$$

(M_{12})

$16 \rightarrow (-1)^3 (-16)$

④ $A^{-1} = \frac{1}{\det A} M^T$

↳ adjacent matrix

19 March

Ex. Inverse of 3*3 matrix

Find the Inverse of this matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{bmatrix} \qquad A^{-1} = ?$$

① Det. $A = 1(5*9-6*2) - 2(4*9-6*7) + 3(4*2-5*7) = -36$

② $M_{11} = \begin{bmatrix} 5 & 6 \\ 2 & 9 \end{bmatrix} = 33$, $M_{21} = \begin{bmatrix} 2 & 3 \\ 2 & 9 \end{bmatrix} = 12$

$M_{13} = -27$ $M_{12} = -6$ $M_{23} = -12$ $M_{22} = -12$

$M_{31} = -3$ $M_{32} = -6$ $M_{33} = -3$

Major $M_{ij} = (-1)^{i+j} M_{ij(minor)}$

→ Major cofactors $= \begin{bmatrix} 33 & 6 & -27 \\ -12 & -12 & 12 \\ -3 & 6 & -3 \end{bmatrix}$

$$A^{-1} = \frac{1}{\det A} A_{adjacent}$$

the transpose of the major cofactors matrix

$$[A^{-1}] = \frac{1}{-36} \begin{bmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{bmatrix}$$

Linear Algebric Eqns.

19 March

Method of solution:

1. Inverse method

$$a_{11}X_1 + a_{12}X_2 = b_1$$
$$a_{21}X_1 + a_{22}X_2 = b_2$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}[B]$$

Example: $3X_1 + 2X_2 = 18$

$-X_1 + 2X_2 = 2$

$$\begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 18 \\ 2 \end{bmatrix} \qquad A^{-1} = ?$$

Solution: $\det A = 8$

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}$$

$X = A^{-1}[B]$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 18 \\ 2 \end{bmatrix}$$

$$X_1 = \frac{2*18 - 2*2}{8} = 4$$

$$X_2 = \frac{1*18 + 3*2}{8} = 3$$

Homework

19 March

Example: Solve the System of linear Eqns. using the Inverse method.

$$0.3X_1 + 0.52X_2 + X_3 = -0.01$$
$$0.5X_1 + X_2 - 1.9X_3 = 0.67$$
$$0.1X_1 + 0.3X_2 + 0.5X_3 = -0.441$$

Solution:

$$\begin{bmatrix} 0.3 & 0.52 & 1 \\ 0.5 & 1 & -1.9 \\ 0.1 & 0.3 & 0.5 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} -0.01 \\ 0.67 \\ -0.441 \end{bmatrix}$$

A X B

$$\det.A = 0.3((1*0.5)-(-1.9*0.3)) - 0.52((0.5*0.5)-(-1.9*0.1)) + 1((0.5*0.3)-(1*0.1))$$

$$\det.A = 0.1422$$

$M_{11} = 1.07$ $M_{12} = 0.44$ $M_{13} = 0.05$

$M_{21} = -0.04$ $M_{22} = 0.05$ $M_{23} = 0.038$

$M_{31} = -1.988$ $M_{32} = -1.07$ $M_{33} = 0.041$

Major $M_{ij} = (-1)^{i+j} M_{ij}$ (minor)

$$\text{Major cofactors} = \begin{bmatrix} 1.07 & -0.44 & 0.05 \\ 0.04 & 0.05 & -0.038 \\ -1.988 & 1.07 & 0.04 \end{bmatrix}$$

$$\text{Adjacent} = \begin{bmatrix} 1.07 & 0.04 & -1.988 \\ -0.44 & 0.05 & 1.07 \\ 0.05 & -0.038 & 0.04 \end{bmatrix}$$

(transposed)

$$A^{-1} = \frac{1}{0.1422} \begin{bmatrix} 1.07 & 0.04 & -1.988 \\ -0.44 & 0.05 & 1.07 \\ 0.05 & -0.038 & 0.04 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 7.53 & 0.28 & -13.97 \\ -3.09 & 0.35 & 7.52 \\ 0.35 & -0.267 & 0.28 \end{bmatrix}$$

19 March

$$X = A^{-1} B$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 7.53 & 0.28 & -13.97 \\ -3.09 & 0.35 & 7.52 \\ 0.35 & -0.267 & 0.28 \end{bmatrix} \begin{bmatrix} -0.01 \\ 0.67 \\ -0.441 \end{bmatrix}$$

$$X_1 = (7.53 * -0.01) + (0.28 * 0.67) + (13.97 * 0.441)$$

$X_1 = 6.2591$

$X_2 = -3.0434$

$X_3 = 0.3056$

19 March

Cramer's Method

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$x_1 = \frac{\begin{bmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{bmatrix}}{\det. A} \qquad x_2 = \frac{\begin{bmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{bmatrix}}{\det. A}$$

Example: $3x_1 + 2x_2 = 18$

$-x_1 + 2x_2 = 2$

$$\begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 18 \\ 2 \end{bmatrix}$$

$\det. A = 8$

$$x_1 = \frac{\begin{vmatrix} 18 & 2 \\ 2 & 2 \end{vmatrix}}{8} = 4$$

$$x_2 = \frac{\begin{vmatrix} 3 & 18 \\ -1 & 2 \end{vmatrix}}{8} = 3$$

19 March

In General

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{\det A}$$

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}}{\det A}$$

$$x_3 = \frac{\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}}{\det A}$$

Lecture 8 - Gauss Elemenation

24 March

$$\underset{3*3}{A}\ \underset{3*1}{X} = \underset{3*1}{B}$$

Augmented matrix of A

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right] \xrightarrow[\text{operations}]{\text{Row}} \left[\begin{array}{ccc|c} c_1 & c_2 & c_3 & b_1' \\ 0 & c_4 & c_5 & b_2' \\ 0 & 0 & c_6 & b_3' \end{array}\right]$$

$$c_6 x_3 = b_3'$$
$$c_4 x_2 + c_5 x_3 = b_2'$$
$$c_1 x_1 + c_2 x_2 + c_3 x_3 = b_3'$$

Example: Use G.E to solve:

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

$$\left[\begin{array}{ccc|c} 3 & -0.1 & -0.2 & 7.85 \\ 0.1 & 7 & -0.3 & -19.3 \\ 0.3 & -0.2 & 10 & 71.4 \end{array}\right]$$

السهم الخارج من العدد يدل على الربط بالعدد
الذي سوف نستخدمه لتصفير العدد
عندما يتم العدد ← يتغير الصف الذي هو فيه كاملاً

24 march

3	-0.1	-0.2	7.85
0.1	7	-0.3	-19.3
0.3	-0.2	10	71.4

* الصف الأول لن يجري عليه أي تغيير

$R_2 \rightarrow R_2 + \left(\frac{-0.1}{3}\right) R_1 \rightarrow 0.1 + \left(\frac{-0.1}{3}\right) * 3 \rightarrow R_2 = 0$

↳ Row 2 ↳ الصف الأول

$7 + \left(\frac{-0.1}{3}\right) * -0.1 = 7.0033 \qquad -0.3 + \left(\frac{-0.1}{3}\right) * -0.2 = -0.2933$

$-19.3 + \left(\frac{-0.1}{3}\right) * 7.85 = -19.5617$

3	-0.1	-0.2	7.85
0	7.0033	-0.2933	-19.5617
0.3	-0.2	10	71.4

$R_3 \rightarrow R_3 + (-0.1) * R_1$

3	-0.1	-0.2	7.85
0	7.0033	-0.2933	-19.5617
0	-0.19	10.02	7.615

$R_3 \rightarrow R_3 + \left(\frac{0.19}{7.0033}\right) * R_2$

$$\left[\begin{array}{ccc|c} 3 & -0.1 & -0.2 & 7.85 \\ 0 & 7.0033 & -0.2933 & -19.5617 \\ 0 & 0 & 10.01204 & 70.0843 \end{array}\right]$$

$C_6 X_3 = b_3'$

$10.01204 X_3 = 70.0843$

$X_3 = 7$

$C_4 X_2 + C_5 X_3 = b_2' \rightarrow X_2 = 7.5$

$C_1 X_1 + C_2 X_2 + C_3 X_3 = b_1' \rightarrow X_1 = 3$

24 March

Gauss Seidel Method

$$a_{11}X_1 + a_{12}X_2 + a_{13}X_3 = b_1$$
$$a_{21}X_1 + a_{22}X_2 + a_{23}X_3 = b_2$$
$$a_{31}X_1 + a_{32}X_2 + a_{33}X_3 = b_3$$

Rewrite the eqns in the form

$X_1 = f(X_2, X_3)$ → (X_1) موضوع القانون

$X_2 = f(X_1, X_3)$

$X_3 = f(X_1, X_2)$

$$\rightarrow X_1 = \frac{1}{a_{11}}\left(b_1 - a_{12}X_2 - a_{13}X_3\right)$$

$$\rightarrow X_2 = \frac{1}{a_{22}}\left(b_2 - a_{21}X_1 - a_{23}X_3\right)$$

$$\rightarrow X_3 = \frac{1}{a_{33}}\left(b_3 - a_{32}X_2 - a_{31}X_1\right)$$

Cheak

$$a_{11} > |a_{12}| + |a_{13}|$$
$$a_{22} > |a_{21}| + |a_{23}|$$
$$a_{33} > |a_{31}| + |a_{32}|$$

(divergent) في حال لم تتحقق الشروط سوف تتباعد الحلول

Start with $X_1 = 0$, $X_2 = 0$, $X_3 = 0$

Then find the new values & Substitude it directly

24 March

Example: $45X_1 + 2X_2 + 3X_3 = 58$

$-3X_1 + 22X_2 + 2X_3 = 47$

$5X_1 + X_2 + 20X_3 = 67$

1st iteration ① $X_1 = 0 \quad X_2 = 0 \quad X_3 = 0$

$\rightarrow X_1 = \frac{1}{45}(58 - 2X_2 - 3X_3) \rightarrow X_1 = 1.2889$

$\rightarrow X_2 = \frac{1}{22}(47 + 3X_1 - 2X_3) \rightarrow X_2 = 2.31212$ ②

$\rightarrow X_3 = \frac{1}{20}(67 - 5X_1 - X_2) \rightarrow X_3 = 2.912127$

لإيجاد قيم المجاهيل الأولية
(3rd iteration) لحل

$X_1 = \frac{1}{45}(58 - 2(2.31212) - 3(2.912127))$

$X_1 = 0.992$

$\rightarrow X_2 = 2.007 \qquad \rightarrow X_3 = 3.0017$

lecture 9 - Jacobi Method

26 March

- The calculated Value of X_1, X_2 & X_3 are used in the next iteration

Example: $45X_1 + 2X_2 + 3X_3 = 58$

$-3X_1 + 22X_2 + 2X_3 = 47$

$5X_1 + X_2 + 20X_3 = 67$

Solution :-

iteration (1) $X_1 = 0$ $X_2 = 0$ $X_3 = 0$

$$X_1 = \frac{1}{45}\left(58 - 2(0) - 3(0)\right) = 1.2889$$

(X_2, X_3)

$$X_2 = \frac{1}{22}\left(47 + 3(0) - 2(0)\right) = 2.1363$$

(X_1, X_3)

$$X_3 = \frac{1}{20}\left(67 - 5(0) - (0)\right) = 3.350$$

(X_1, X_2)

في أول iteration نفرض Zero X_3, X_2, X_1

هذا هو الاختلاف بين Jacobi & Seidel

iteration (2)

$$X_1 = \frac{1}{45}\left(58 - 2(2.1363) - 3(3.350)\right) = 0.9706$$

$$X_2 = \frac{1}{22}\left(47 + 3(1.2889) - 2(3.350)\right) = 2.008$$

$$X_3 = \frac{1}{20}\left(67 - 5(1.2889) - 2.136\right) = 2.9209$$

→

26 March

iteration No.3

$$X_1 = \frac{1}{45}(58 - 2(2.008) - 3(2.9209)) = 1.0049$$

$$X_2 = \frac{1}{22}(47 + 3(0.9706) - 2(2.9209)) = 2.0031$$

$$X_3 = \frac{1}{20}(67 - 5(0.9706) - 2.008) = 3.0069$$

⋮

$X_1 = 1.0$

$X_2 = 2.0$

$X_3 = 3.0$

Relative % Error for X_1

$$\frac{X_1^{New} - X_1^{old}}{X_1^{New}} * 100\ \% \rightarrow$$

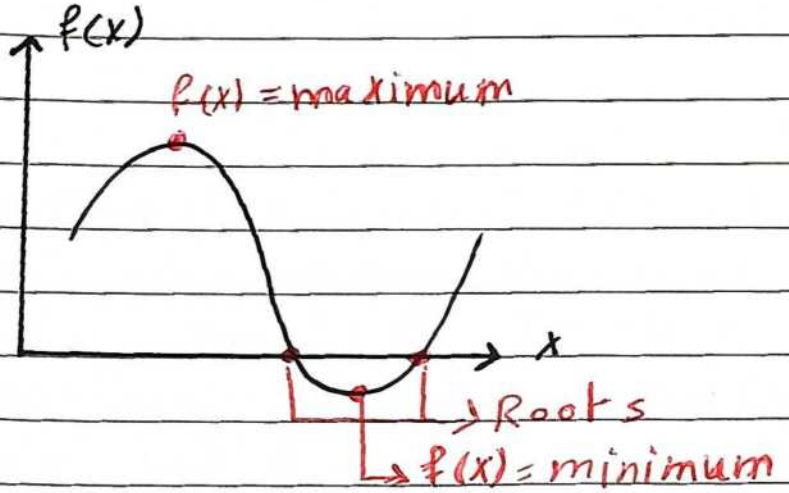
بعد ما نوجد الـ X_1 المطلوبة
في الـ iteration المطلوبة
بعدين يتم تطبيق القانون على X_2 و X_3

optimization lectur 1o — 7 April

to find the optimum values of a fn

- one dimensional: fn → f(x)
- two dimensional: fn → f(x, y)

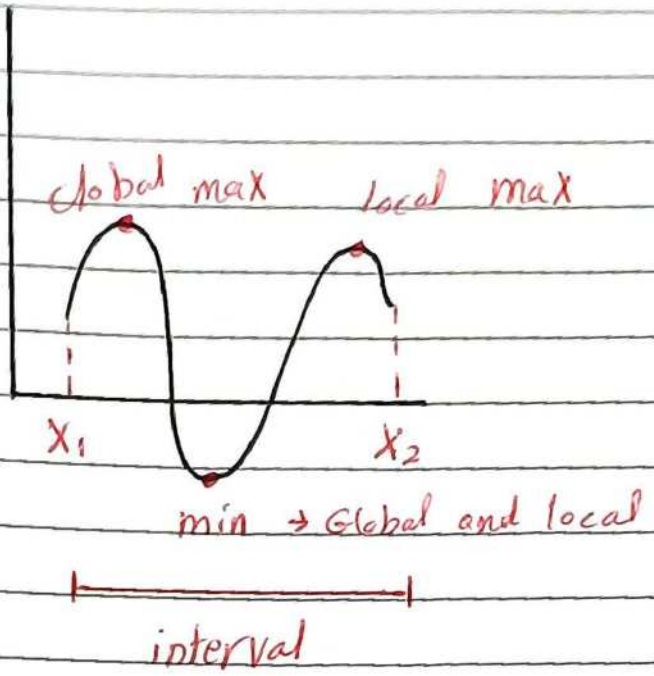
Optimum
- maximum
- minimum



for maximum $f'(x) = 0$ & $f''(x) < 0$

for minimum $f'(x) = 0$ & $f''(x) > 0$

7 April



Within this interval ($X_1 \rightarrow X_2$) the function has two maximum values i.e this function is a multimodal fn, IF the fn has one maximum or minimum within an interval i.e this function is a unimodal

7 April

one dimensional:

1. Newton's method
2. Golden Search section method (GSS)

two dimensional:

1. Gradient method
2. steepest method

Newton's method:

Algorithm: start with an initial value of X. X_0 (initial Guess)

$$X_{i+1} = X_i - \frac{f'(X_i)}{f''(X_i)}$$

$$X_1 = X_0 - \frac{f'(X_0)}{f''(X_0)} \qquad X_2 = X_1 - \frac{f'(X_1)}{f''(X_1)}$$

$$Error = \frac{X_{new} - X_{old}}{X_{new}} * 100\%$$

يتوقف عندما تصبح $f'(x)$ تساوي zero

7 April

Example: find the maximum of the fn, $f(x) = 2\sin x - \frac{x^2}{10}$, start with $x_0 = 2.5$

Solution:

$f'(x) = 2\cos x - \frac{x}{5}$

$f''(x) = -2\sin x - \frac{1}{5}$

$x_1 = 2.5 - \frac{f'(2.5)}{f''(2.5)}$

$x_1 = 0.99508$, $f'(2.5) = -2.10229$

$x_2 = 0.99508 - \frac{f'(0.99508)}{f''(0.99508)}$

$x_2 = 1.46901$, $f'(0.99508) = 0.88985$

$x_3 = 1.42764$, $f'(0.88985) = -0.09058$

$x_4 = 1.42755$, $f'(1.42764) = 0.0002 \approx 0$ ※

maximum value of $f(x)$:

$f(1.42755) = 1.77573$ ※

7 April

Gss method :

1. select an interval X_L & X_u (two values), X in between

2. Use a golden number : $\varnothing = 0.61803$

Define the distance : $d = \varnothing (X_u - X_L)$

then compute the two interior Points:

$X_1 = X_L + d$ & $X_2 = X_u - d$

← الهدف تضييق الفترة

3. Evaluate $f(X_1)$ & $f(X_2)$

To search the maximum

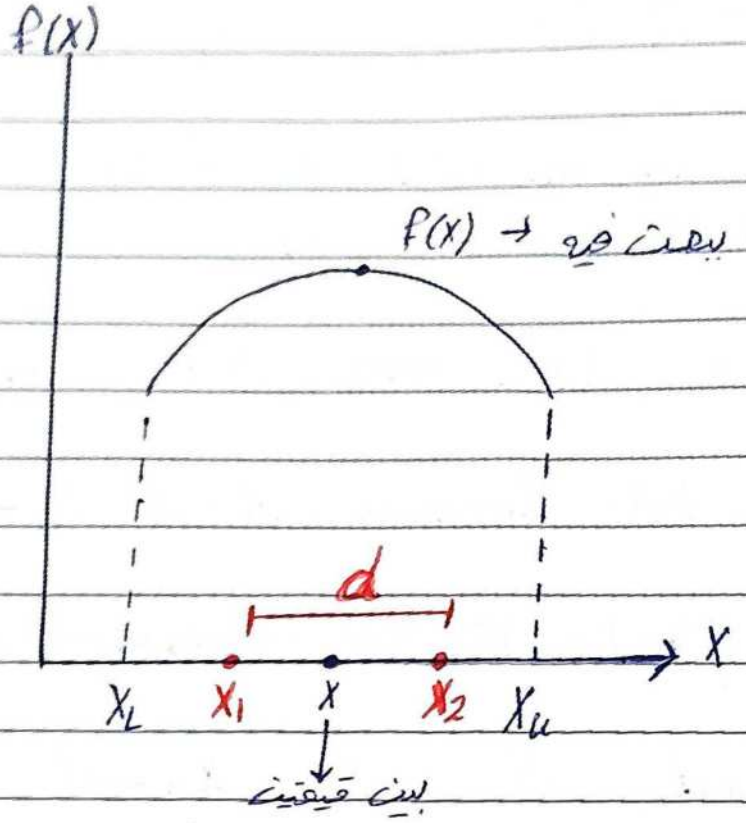
- if $f(X_1) > f(X_2)$ $X_L = X_2$ & $X_u = X_u$
  $f(X_1) < f(X_2)$ $X_u = X_1$ & $X_L = X_L$

To search the minimum

- if $f(X_1) > f(X_2)$ $X_u = X_1$ & $X_L = X_L$
  $f(X_1) < f(X_2)$ $X_L = X_2$ & $X_u = X_u$

→ توضيحات على Gss method

7 April



* بدي اضل احصر القيمة على يمين X والقيمة على يسار X لحد ما أوصل X

اليه الحصر وتصغير الفترة:

$X_1 = X_L + d$

بالتالي X_1 بين X و X_L

$X_2 = X_u - d$

بالتالي X_2 بين X و X_u

وبذلك نقوم بتصغير حتى نوصل X

θ: Golden number

θ is constant = 0.61803

* العدد الذهبي يقسم الخط بحيث يكون الجزء الأكبر إلى الأصغر بنفس النسبة دائماً وهذا يجعل البحث متوازن وسريع

$$d = \theta(X_u - X_L)$$

Lecture 11 — 9 April

Multidimensional optimization
2-D optimization

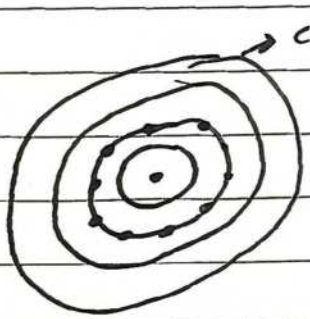
1- Gradient method :

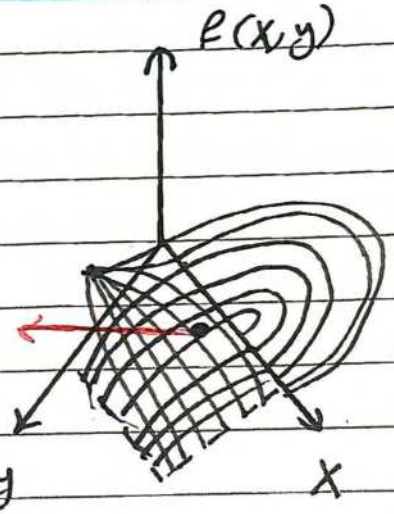
Gradient vector ∇f = $\begin{bmatrix} df/dx_1 \\ df/dx_2 \\ df/dx_3 \\ \vdots \\ df/dx_n \end{bmatrix}$

∇ : ديل

X_2 و X_1 ← مش على نفس الـ axis

one dimentional	two dimentional
$f_n = f(x)$	$f_n = f(x,y)$





All Points have the same $f(x,y)$ value

9 APril

one dimentional

$df/dx = 0$ & $\frac{d^2f}{dx^2} < 0$ maximum

$df/dx = 0$ & $\frac{d^2f}{dx^2} > 0$ minimum

- يتم تحديد نوع النقطة الحرجة فقط بإستخدام المشتقة الثانية

Two dimenbional

$df/dx = 0$ & $\frac{d^2f}{dx^2} < 0$ maximum !?

$df/dx = 0$ & $\frac{d^2f}{dx^2} > 0$ minimum !?

المشتقات الجزئية الأولى تعطينا النقاط الحرجة
لكن لتعرف نوع النقطة (صغرى/عظمى/سرجية) ، المشتقة الثانية لوحدها غير كافية ، بسبب وجود اكثر من اتجاه ممكن للدالة أن تنحني فيه (x & y) لذلك نحتاج :

(Hessian matrix)

Use Hessian matrix to judge whether it is a maximum or minimum or Saddle

لـ سرج

9 April

Heissian matrix

$$\begin{bmatrix} d^2f/dx^2 & d^2f/dxdy \\ d^2f/dydx & d^2f/dy^2 \end{bmatrix}$$

$$\det.|H| = (d^2f/dx^2)(d^2f/dy^2) - (d^2f/dxdy)^2$$

if $df/dx = 0$, $df/dy = 0$, $d^2f/dx^2 < 0$

& $|H| > 0$: maximum

$df/dx = 0$, $df/dy = 0$, $d^2f/dx^2 > 0$

& $|H| > 0$: minimum

if $|H| < 0$: Sadde Point (SP)

Example:

For the fn: $f(x,y) = 2xy + 1.5y - 1.25x^2 - 2y^2$
what is the point which the fn value is maximum or minimum or saddle

Solution:

$$\nabla f = 0 = \begin{bmatrix} df/dx \\ df/dy \end{bmatrix}$$

$df/dx = 2y - 2.5x = 0$ ---①
$df/dy = 2x + 1.5 - 4y = 0$ ---②
Solve for: $x = 0.5$ $y = 0.625$

$d^2f/dx^2 = -2.5 < 0$
$d^2f/dy^2 = -4.0$

$$d^2f/dxdy = \frac{d}{dy}\cdot\frac{df}{dx} = \frac{d}{dy}(2y - 2.5x) = \boxed{2}$$

$$[H] = \begin{bmatrix} -2.5 & 2 \\ 2 & -4 \end{bmatrix}$$

$|H| = 6 > 0$ →

9 April

$|H| > 0$

$d^2P/dx^2 = -2.5 < 0$: maximum

Point $(0.5, 0.625)$ ✻

قيمة max ← $P(x,y) = 2(0.5)(0.625) + 1.5(0.625) - 1.25(0.5)^2 - 2(0.625)^2$

[التعويض في المعادلة $P(x,y)$ ✻]

$= 0.46875$ ✻ قيمة max.

9 April

2- Optimal Steepest Ascent method:

Algorithm:

1. Select an initial point (x_i, y_i)

2. Find $\left[\frac{df}{dx}\right](x_i, y_i)$ & $\left[\frac{df}{dy}\right](x_i, y_i)$
   - $\left[\frac{df}{dx}\right] \rightarrow \nabla f_x$
   - $\left[\frac{df}{dy}\right] \rightarrow \nabla f_y$

3. New x and y values

$$x_{i+1} = x_i + \nabla f_x(h)$$

$$y_{i+1} = y_i + \nabla f_y(h)$$

4. Substitute thes values in $f(x_{i+1}, y_{i+1})$ to call fn $g(x)$

5. let $g'(x) = 0$ to find h to determine the next point

9 April

Example: Maximize the
fn: $f(x,y) = 2xy + 2x - x^2 - 2y^2$
use the optimum (steepest) Ascent method (OSA method)
to find the next point if the initial
point is $(-1, 1)$.

Solution:

$$\partial f/\partial x = 2y + 2 - 2x \qquad \partial f/\partial y = 2x - 4y$$

$$\nabla f_{(-1,1)} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$$

$$g(x) = f(-1+6h, 1-6h)$$

بتعويض بالاقتران $f(x,y)$ ⟵ $f(-1+6h, 1-6h) = -7 + 72h - 180h^2 = g(x)$

$$g'(x) = 72 - 360h = 0$$

$$h = 0.2$$

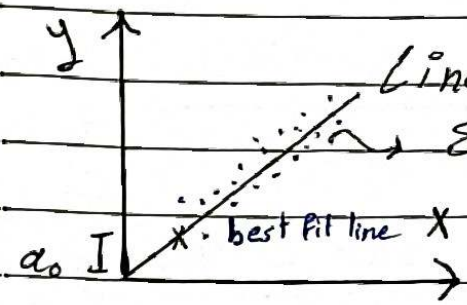
$$x_{i+1} = -1 + 6(0.2) = 0.2$$
$$y_{i+1} = 1 - 6(0.2) = -0.2$$

the two next points في حال طلب
بعد ايجاد $(0.2, -0.2)$ ← بعتبرها initial بالسؤال ويرجع
بعمل السؤال بتلك القيم الجديدة.

Curve fitting (Regression)

14 April

for sets of x & y values $(x_1, y_1), (x_2, y_2), (x_3, y_3)$



Polynomial (2nd degree)

$$y = a_0 + a_1 x + a_2 x^2$$

in general: $y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$

To determine the best fit curve calculate the standard deviation S and the correlation coefficient $= R^2$

Linear $R^2 = 0.8692$ $S = 1.72$
Polynomial $R^2 = 0.9725$ $S = 1.25$

$R^2 = 0.8692 \rightarrow 86.92\%$ of uncertainity is removed

$R^2 = 0.9725 \rightarrow 97.25\%$ of uncertainity is removed

$\rightarrow 2.75\%$ Error

14 April

Linear Regression

$$a_1 = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

* n = number of sets standard Error.

$$a_0 = \frac{\Sigma y_i - a_1 \Sigma x_i}{n} = \bar{y} - a_1 \bar{x}$$

Standard deviation: يقيس تشتت القيم حول متوسطها $S = \sqrt{\frac{\Sigma (y_i - \bar{y})^2}{n-1}}$

Standard Error: يقيس مدى تشتت النقاط حول خط الإنحدار $S_{y/x} = \sqrt{\frac{S_r}{n-2}}$

$S_r = \Sigma (y_i - a_0 - a_1 x_i)^2$ يعبر على مقدار التشتت او الانحراف من القيم الفعلية عن القيم المتوقعة

Example: Find the linear B.F. curve for the following sets:

x_i	y_i	$x_i y_i$	$(x_i)^2$	$(y_i - \bar{y})^2$	$(y_i - a_0 - a_1 x_i)^2$
1	0.5	0.5	1	8.579	0.1687
2	2.5	5	4	0.863	0.5625
3	2.0	6	9	2.042	0.3473
4	4.0	16	16	0.326	0.3265
5	3.5	17.5	25	0.005041	0.5897
6	6	36	36	6.610	0.7970
7	5.5	38.5	49	4.289	0.1994

$\Sigma x_i = 28$ $\Sigma y_i = 24$ $\Sigma x_i y_i = 119.5$ $\Sigma x_i^2 = 140$

$\bar{x} = \frac{28}{7}$ $\bar{y} = \frac{24}{7}$

$\bar{x} = 4$ $\bar{y} = 3.429$

$\Sigma (y_i - \bar{y})^2 = 22.7143$

$S_r = \Sigma = 2.9911$ →

$S_{y/x} = 0.7734$

14 April

$$a_1 = \frac{7(119.5) - (28)(24)}{7(140) - (28)^2} = 0.8393$$

$$a_0 = \frac{24 - 0.8393(28)}{7} = 0.07143$$

$y = a_0 + a_1 x \rightarrow 0.07143 + 0.8393x$

$R^2 = 0.868$ $\quad R^2 = \frac{\Sigma(y_i - \bar{y})^2 - S_r}{\Sigma(y_i - \bar{y})^2}$

↳ correlation coefficient

$S_{y/x} = 0.7735$ $\quad S = 1.9457$

R^2: coefficient of determination

R: correlation coefficient

2) Polynomial Regression

$$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 & \cdots \\ & & & & \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 & \cdots & \\ & & & & \\ & \Sigma x_i^4 & & & \\ & & & & \\ \text{Symmetrical} & & \Sigma x_i^{2m} & & \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_3 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \Sigma x_i^2 y_i \\ \vdots \\ \Sigma x_i^m y_i \end{bmatrix}$$

n: No. of sets
m: order

$$\text{Standard Error} = \sqrt{\frac{S_r}{n-(m+1)}}$$

$$S_r = \Sigma (y_i - a_0 - a_1 x_i \cdots)^2$$

$$R^2 = \left(\Sigma (y_i - \bar{y})^2 - S_r\right) / \Sigma (y_i - \bar{y})^2$$

Example: Find the 2nd order Polynomial B.F.C for the following sets.

x_i	y_i
0	21
1	7.7
2	13.0
3	27.2
4	40.9
5	61.1

Solution

$n = 6$

$m = 2$

$\bar{x} = 2.5$

$\bar{y} = 25.433$

$\Sigma x_i^4 = 979$

$\Sigma x_i y_i = 585.6$

$\Sigma x_i^2 y_i = 2488.8$

$\Sigma x_i = 15$

$\Sigma y_i = 152.6$

$\Sigma x_i^2 = 55$

$\Sigma x_i^3 = 225$ →

14 April

6	15	55	a_0		152.6
15	55	225	a_1	=	585.6
55	225	979	a_3		2488.8

Solve for $a_0 = 2.4786$, $a_1 = 2.3593$,
$a_2 = 1.8607$

$y = 2.4786 + 2.3593x + 1.8607x^2$

$S_{y/x} = 1.12$ $R^2 = 0.99851$

Interpolation

16 April

is to find a value between 2 or more values.

X	$y = f(x)$
1	2.75
2	3.49
3	4.92

$f(2.12) = ?$

Extrapolation: is to find the value outside 2 or more points

	X	$f(x)$
x_0	1	2.75 b_0
x_1	2	3.49
x_2	3	4.92

$f(4.07) = ?$

Interpolation method solution:

1. Newton's divided difference method (NDD)
2. Lagrange ~~Interpolation~~ Method (LM)

16 April

NDD Method

For 2 Points → 1st order $P(x) = b_0 + b_1(x - x_0)$

3 Points → 2nd order $P(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$

4 Points → 3rd order $P(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$

$b_0 = y(x_0)$

$b_1, b_2, b_3, \ldots$ are the slopes between Points

Example

Estimate $\ln(2)$ between $\ln(1) = 0$, $\ln(6) = 1.791759$

↳ الهدف هنا → (Error) هي القيمة الحقيقية وبدنا منها الـ

1st order → لأنه بشتغل على نقطتين

↳ في حال مش موجودة فا بدنا الـ Error

	x_i	y_i	b_i
x_0	1	0	0.3583518
x	2		
x_1	6	1.791759	

$b_0 = y_0 =$ zero

$b_1 = \frac{1.791759 - 0}{6 - 1} =$

b_1

$P(x) = b_0 + b_1(x - x_0)$

$= 0 + (0.3583518)(2 - 1)$

$= 0.3583518$

$\text{Error} = \frac{0.6931472 - 0.3583518}{0.6931472} \times 100\%$

$E_t = 48.3\%$

16 April

Example :

Find $P(2)$ if $P(1) = \ln(1) = 0$

$P(4) = \ln(4) = 1.386294$

$P(6) = \ln(6) = 1.791759$

	x_i	y_i	b_1	b_2
x_0	1	0	0.462098	-0.051873
x_1	4	1.386294		
x_2	6	1.791759	0.202733	

$\rightarrow \frac{1.386294 - 0}{4 - 1} = 0.462098$ b_1

Note: $b_0, b_1, b_2, \ldots$ للصف الأول من الجدول

$\rightarrow \frac{1.791759 - 1.386294}{6 - 4} = 0.202733$

$\rightarrow \frac{0.202733 - 0.462098}{6 - 1} = -0.051873$

بداية الرقم الأول ← لنهاية الرقم الثاني

$P(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$

$P(x) = 0 + 0.462098(2 - 1) + (-0.051873(2 - 1)(2 - 4)$

$P(2) = 0.565844$

$\varepsilon\% = \frac{0.693147 - 0.565844}{0.693147} \times 100\% = 18.48\%$

16 April

Lagrange Method

Linear version $n = 1$

$$f_1(x) = \left(\frac{x - x_1}{x_0 - x_1}\right) f(x_0) + \left(\frac{x - x_0}{x_1 - x_0}\right) f(x_1)$$

2nd order

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0)$$

$$+ \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_1)(x - x_0)}{(x_2 - x_1)(x_2 - x_0)} f(x_2)$$

Example

find $f(5)$ if $f(2) = 9$, $f(6) = 17$ using (L.M)

Solution

$x_0 = 2$, $f(x_0) = 9$

$x_1 = 6$, $f(x_1) = 17$

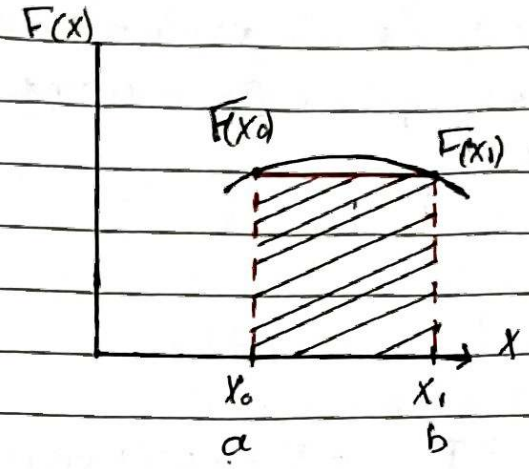
$x = 5$, $f(5) = ??$

For 2 Points problem use Linear version → $f_1(5) = \left(\frac{5-6}{2-6}\right)(9) + \left(\frac{5-2}{6-2}\right)(17)$

$f_1(5) = 15$ ✓

Numerical Evaluation of integration

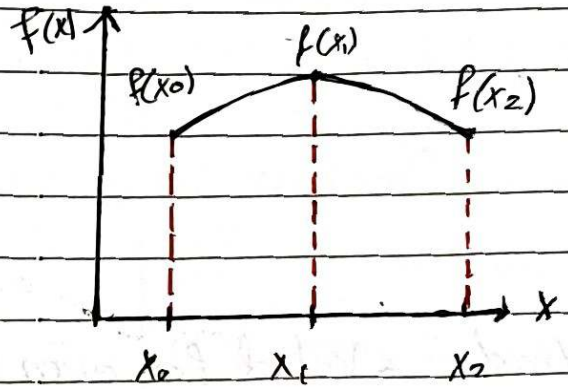
21 April



$\int_{x_0}^{x_1} f(x)\,dx$ = area under the fn. line.

Trapezoidal Method

$$I = (X_1 - X_0)\left(\frac{f_{x_0} + f_{x_1}}{2}\right)$$



$$I = h\frac{f(x_0)+f(x_1)}{2} + h\frac{f(x_1)+f(x_2)}{2}$$

$$= \frac{h}{2}\left(f(x_0) + 2f(x_1) + f(x_2)\right)$$

In General:

$$I = \frac{h}{2}\left(f(x_0) + 2(f(x_1) + f(x_2) + \dots + f(x_{n-1})) + f(x_n)\right)$$

21 April

Example: $\int_0^3 \frac{x^4+2}{x+1}\,dx$ Evaluate this Fn. using $h=0.5$

$$F(x) = \frac{x^4+2}{x+1}$$

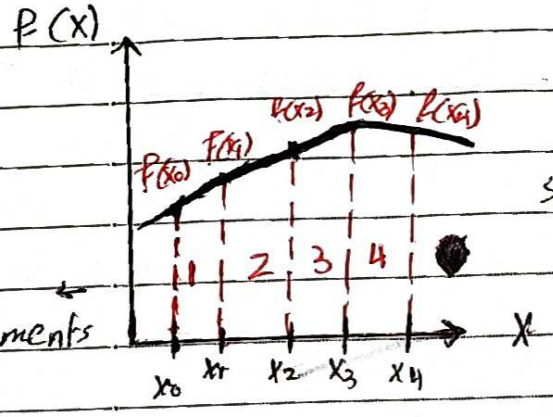
x_i	$f(x_i)$
0	2.000
0.5	1.375
1	1.500
1.5	2.825
2	6.000
2.5	11.730
3	20.750

$$I = \frac{0.5}{2}(2+2(1.375+1.5+2.825+6+11.730)+20.75)$$

$$I = 17.4025$$

Simpson's Rule

1) 1/3 Simpson's method * Valid for even No. of segments.



The No. of segments is (4)

$$I = \frac{h}{3}\left(f(x_0) + f(x_n) + 4\sum_{\text{Even}} f(x_i) + 2\sum_{\text{odd}} f(x_i)\right)$$

Solution:

$$I = \frac{h}{3}\left(f(x_0) + f(x_4) + 4(f(x_1)+f(x_3)) + 2(f(x_2))\right)$$

21 April

Example: Evaluate this fn. using $h = 0.5$ by Simpson's 1/3 method.

$$\int_0^3 \frac{x^4+2}{x+1}$$

X_i	$f(x)$
0	2
0.5	1.375
1	1.500
1.5	2.825
2	6.000
2.5	11.730
3	20.750
3.5	33.791

Solution:

$I = I_{1-6} + I_7$

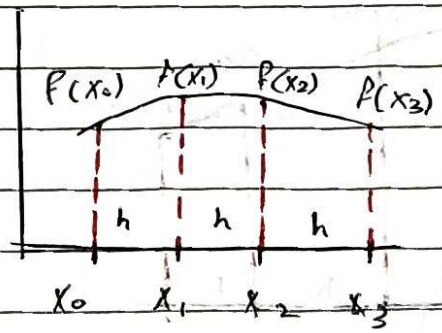
$I_{1-6} = \frac{0.5}{3}(2 + 20.750 + 4(1.375 + 2.825 + 11.730) + 2(1.5+6))$

$I_{1-6} = 16.911$

$I_7 = \frac{0.5}{2}(20.750 + 33.791) = 13.635$

$I = 16.911 + 13.635 = 30.546$

* 3/8 simpson's method:



Valid for 3 segments or multiples of 3

$$I = \frac{3}{8} h \left(f(x_0) + f(x_3) + 3(f(x_1) + f(x_2)) \right)$$

21 April

Example:

x_i	$f(x_i)$
0	2.000
0.5	1.375
1	1.500
1.5	2.825

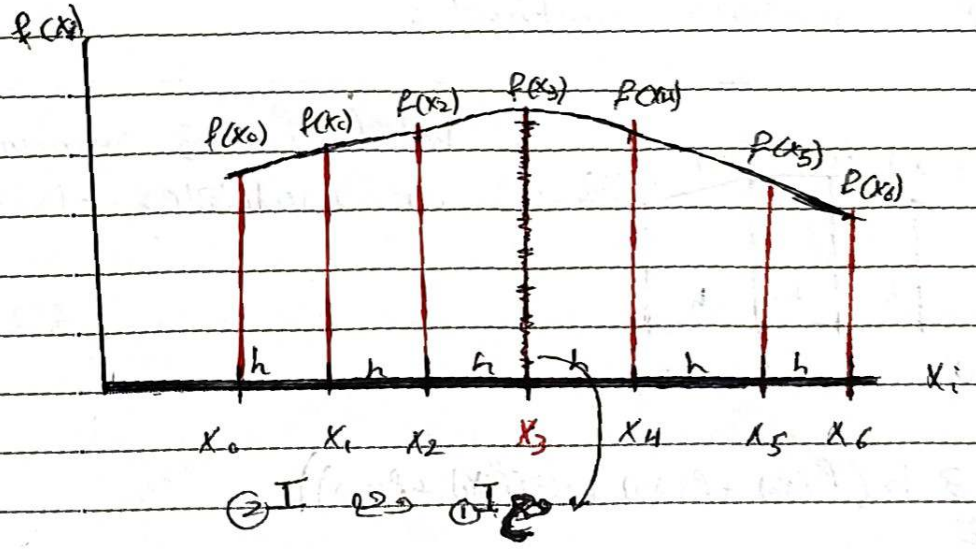
Solution: $I = \frac{3}{8}(0.5)(2 + 2.825 + 3(1.375 + 1.5))$

$I = 2.521875$

إذا كان السؤال 6-segments

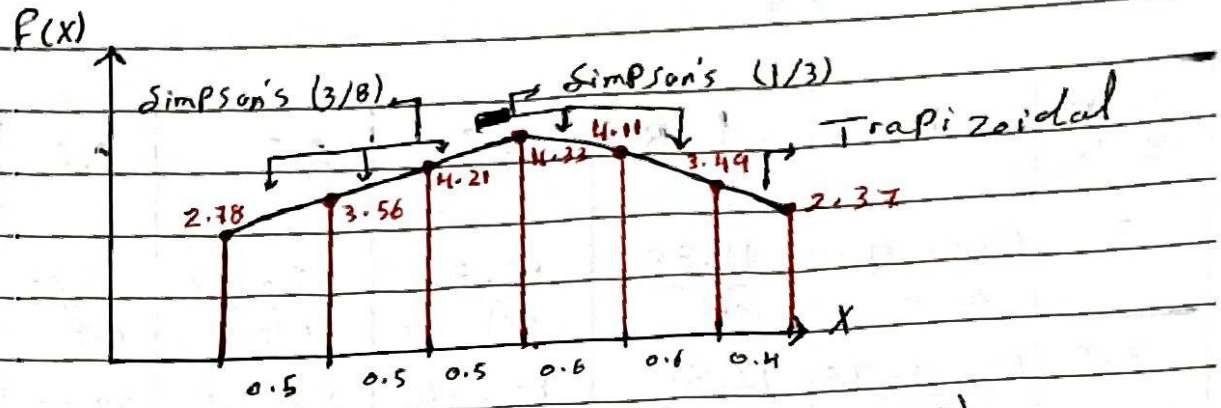
يصل العلاقة على انها جزئين كالتالي:

$$\rightarrow I = \frac{3}{8}h\left(f(x_0) + f(x_6) + 2f(x_3) + 3(f(x_1) + f(x_2) + f(x_4) + f(x_5))\right)$$



Unequal segments

23 April



$$I_1 = \frac{3}{8}(0.5)(2.78 + 4.33 + 3(3.56 + 4.21))$$

$$I_2 = \frac{1}{3}(0.6)(4.33 + 3.49 + 4(4.11))$$

$$I_3 = \frac{0.4}{2}(3.49 + 2.37)$$

* Numerical Evaluation of the double Integration's

$$\int_a^b \int_c^d F(x,y)\,dx\,dy$$

* in the X-direction the segment length is $\boxed{h}$
* in the y-direction the segment length is $\boxed{k}$

23 Nisan

$$\int_1^3 \int_3^5 \frac{x^3y^2 + 6xy}{xy}\, dx\, dy \qquad h = 0.4 \qquad k = 0.5$$

	x / y	1	1.4	1.8	2.2	2.6	3
I_1 →	3	F(1,3) = 9.0 corner	11.88 ext.	15.72 ext.	20.52 ext.	26.28 ext.	33 corner
I_2 →	3.5	F(1,3.5) = 9.5 ext.	12.86 int.	17.34 int.	22.94 int.	29.66 int.	37.5 ext.
I_3 →	4	F(1,4) = 10.0 ext.	13.84 int.	18.96 int.	25.36 int.	33.04 int.	42 ext.
I_4 →	4.5	F(1,4.5) = 6.5 ext.	14.82 int.	20.58 int.	27.78 int.	36.42 int.	46.5 ext.
I_5 →	5	F(1,5) = 11.0 corner	15.80 ext.	22.2 ext.	30.2 ext.	39.8 ext.	51 corner

$$I = \frac{hk}{4}\left(\sum F_{\text{corner values}} + 2\sum F_{\text{ext. values}} + 4\sum F_{\text{interior values}}\right)$$

$$\sum F_{corner} = 9.0 + 11 + 33 + 51 = 104$$

$$\sum F_{ext} = 11.88 + 15.72 + 20.52 + 26.28 + 15.8 + 22.2 + 30.2 + 39.8 + 9.5 + 10 + 10.5 + 37.5 + 42 + 46.5 = 338.4$$

$$\sum F_{int.} = 12.86 + 13.84 + 14.82 + 17.34 + 18.96 + 20.58 + 22.94 + 25.36 + 27.78 + 29.66 + 33.04 + 36.42 = 273.6$$

$$I = 93.76$$

23 April

→ X-direction $h = 0.4$

$$I_1 = \frac{0.4}{2}(9 + 33 + 2(11.88 + 15.72 + 20.52 + 26.28)) = A_1$$

$$\vdots$$

$$I_5 = A_5$$

I_5, I_4, I_3, I_2, I_1 بنوجد

$A_1 = 38.16$ $A_2 = 42.52$ $A_3 = 46.88$

$A_4 = 51.24$ $A_5 = 55.6$

$$I = \frac{0.5}{2}(A_1 + A_5 + 2(A_2 + A_3 + A_4))$$

$$I = 93.76$$ ✗

Integration 28 April

Romberg method

	1st order	2nd order	3rd order	- - -
R(1,0)	—	—	—	—
R(2,0)	R(2,1)	—	—	—
R(3,0)	R(3,1)	R(3,2)	—	—
R(4,0)	R(4,1)	R(4,2)	R(4,3)	—
⋮	⋮	⋮	⋮	—

For 4th order $\int F(x)\,dx = R(5,4)$

For 3rd order $\int F(x)\,dx = R(4,3)$

R(1,0), R(2,0), ..., R(4,0) & so on are calculated using trapezoidal Rule For different h value.

For example: (R(1,0); h=1), (R(2,0); h=1/2) (R(3,0); h=1/4), (R(4,0); h=1/8)

→ for other value R(i,k):

$$R(i,k) = \frac{4^k R(i,k-1) - R(i-1,k-1)}{4^k - 1}$$

i ≥ 1, k ≥ 1

28 April

Example: use R.m to evaluate $\int_1^2 \frac{2x^3+3x^2+1}{2}\,dx$ 2nd order

	1st order	2nd order
R(1,0)		
R(2,0)	R(2,1)	
R(3,0)	R(3,1)	R(3,2)

→ For $h=1$ $f(1)=3$ $f(2)=14.5$

$R(1,0) = \frac{1}{2}(3+14.5) = \boxed{8.75}$

→ For $h=1/2$ $f(1)=3$, $f(1.5)=7.25$, $f(2)=14.5$

$R(2,0) = \frac{0.5}{2}(3+14.5+2(7.25)) = \boxed{8}$

→ For $h=1/4$ $f(1)=3$, $f(1.25)=4.797$, $f(1.5)=7.25$,

$f(1.75)=10.453$, $f(2)=14.5$

$R(3,0) = \frac{0.25}{2}(3+14.5+2(4.797+7.25+10.453))$

$R(3,0) = \boxed{7.8125}$

→ $R(2,1) = \left(4^1 R(2,1-1) - R(2-1,1-1)\right) / (4)^1 - 1$

$R(2,1) = (4R(2,0) - R(1,0))/3$

Cubic accuracy

$R(2,1) = 7.75$

→ $R(3,1) = 7.75$

→ $R(3,2) = \left(4^2 R(3,2-1) - R(3-1,2-1)\right)/(4)^2 - 1$

$= (16R(3,1) - R(2,1))/15$

$= 7.75$ ✱

$\int_1^2 f(x)\,dx = 7.75 = R(3,2)$

Numerical Differentiation

30 April

Given $f(x)$ → Required $f'(x)$

Method of solution:

1. Forward finite divided difference.
2. Backward " " " .
3. Centered " " " .

The General Form is:

$$f(x_{i+1}) = f(x_i+h) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!}$$

This General form leads to the followings:

Forward $f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$

backward $f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$

centered $f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$

h: is the step size.

30 April

$f(x) = x^3 - 2x^2 + x + 2$, $f'(x) = 3x^2 - 4x + 1$, $h = 0.5$, $x_i = 2$

Exact Value $f'(2) = 3(2)^2 - 4(2) + 1 = 5$

$f(x_i) = f(2) = 4.0$

$f(x_{i+1}) = f(2.5) = 7.625$

$f(x_{i-1}) = f(1.5) = 2.375$

$\varepsilon\% = \left|\frac{5 - 7.25}{5}\right| * 100\%$

Forward: $f'(2) = \frac{7.625 - 4}{0.5} = 7.25$ $\quad \varepsilon\% = 45\%$

backward: $f'(2) = \frac{4 - 2.375}{0.5} = 3.25$ $\quad \varepsilon\% = 35\%$

centered: $f'(2) = \frac{7.625 - 2.375}{2(0.5)} = 5.25$ $\quad \varepsilon\% = 5\%$

* في حال ما طلب طريقة بشتغل على الـ centered
→ is the more accurate.

Ex. Repeat the previous Example use $h = 0.25$

$x_i = 2.0$ $\quad f(x_i) = 4.0$

$x_{i+1} = 2.25$ $\quad f(x_{i+1}) = 5.5156$

$x_{i-1} = 1.75$ $\quad f(x_{i-1}) = 2.9844$

Forward: $f'(2) = \frac{5.5156 - 4}{0.25} = 6.0624$, $\varepsilon\% = 22.24\%$

backward: $f'(2) = \frac{4 - 2.9844}{0.25} = 4.0624$, $\varepsilon\% = 18.75\%$

centered: $f'(2) = \frac{5.5156 - 2.9844}{2(0.25)} = 5.0624$, $\varepsilon\% = 1.25\%$

Richardson Extrapolation

5 may

Special case: $h_2 = h_1/2$

$$D = \frac{4}{3} D_{h_2} - \frac{1}{3} D_{h_1}$$

D: Derivative; it is preferable to use central, because it is more accurate.

Example: $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

Find $f'(0.5)$ using R.E method with $h = 0.5$.

Solution:

The true value is $f'(0.5) = -0.9125$

Central: $f'(x_i) = \dfrac{f(x_{i+1}) - f(x_{i-1})}{2h}$

$h_1 = 0.5$
$x_i = 0.5$
$x_{i+1} = 1$
$x_{i-1} = 0$

$$D = \frac{4}{3} D_{h_2} - \frac{1}{3} D_{h_1}$$

$$f'(0.5) = \frac{f(1) - f(0)}{2(0.5)} = \frac{0.2 - 1.2}{2(0.5)} = -1 \quad \text{※} D_{h_1}$$

($f(1)$: x_{i+1}; $f(0)$: x_{i-1}; $0.5 = h_1$)

$h_2 = 0.25$
$x_i = 0.5$
$x_{i+1} = 0.75$
$x_{i-1} = 0.25$

$$f'(0.5) = \frac{f(0.75) - f(0.25)}{2(h_2)} = \frac{0.6363 - 1.1035}{2(0.25)} = -0.9344 \quad \text{※} D_{h_2}$$

$$D = \frac{4}{3}(-0.9344) - \frac{1}{3}(-1) = -0.9125 \quad \text{※}$$

$$\varepsilon\% = \left| \frac{\text{True} - \text{Approximate}}{\text{True}} \right| \times 100\% = \text{Zero}$$

Centered finite-divided difference formulas:

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} \qquad O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2})}{12h} \qquad O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} \qquad O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2})}{12h^2} \qquad O(h^4)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2})}{2h^3} \qquad O(h^2)$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3})}{8h^3} \qquad O(h^4)$$

Many other formulas can be found in pages 656-658 (Textbook) 7th edition

Derivatives of equally & unequally spaced data 17 May

Given:

x_i	y_i ← value of $f(x)$
x_0	y_0
x_1	y_1
x_2	y_2
x_3	y_3
⋮	⋮

لكن ما عندي اقوان

Find $f'(x)$ or $f''(x)$

solution?

Find $f(x)$ using Newton's method

Near the beginning of the table

Near the end values of the table

x_i	y_i	b_1	b_2	b_3
x_0	y_0	b_{11}	b_{21}	b_{31}
x_1	y_1	b_{21}	b_{22}	$\triangle\ x-x_1$
x_2	y_2	b_{31}	$\bigcirc\ x-x_2$	$\triangle\ x-x_2$
x_3	y_3	$\square\ x-x_3$	$\bigcirc\ x-x_3$	$\triangle\ x-x_3$

Near the beginning:

$$f(x) = \underbrace{b_0}_{y_0} + \underbrace{b_1}_{b_{11}}(x-x_0) + \overbrace{b_2}^{b_{21}}(x-x_0)(x-x_1) + \overbrace{b_3}^{b_{31}}(x-x_0)(x-x_1)(x-x_2)$$

Near the end:

$$f(x) = \underbrace{b_0}_{y_0} + \underbrace{b_1}_{b_{31}}(x-x_3) + \underbrace{b_2}_{b_{22}}(x-x_2)(x-x_3) + \underbrace{b_3}_{b_{31}}(x-x_1)(x-x_2)(x-x_3)$$

الجزء الضعال الغرائي و x_3

→

12 May

Near the beginning = Near the end

Example: find the approximate values of $f'(1.2)$, $f'(7)$, $f''(0)$ using the calculated value:

x_i	y_i	b_1	b_2	b_3
2	5	-5	-10	6
4	-5	-35	20	
5	-40	25		
7	[illegible]			

$$f(x) = 5 - 5(x-2) - 10(x-2)(x-4) + 6(x-2)(x-4)(x-5)$$

$$f(x) = 5 - 5x + 10 - 10x^2 + 40x + 20x - 80$$
$$+ 6x^3 - 30x^2 - 24x^2 + 120x - 12x^2 + 60x + 48x - 240$$

$$f(x) = 6x^3 - 76x^2 + 283x - 305$$

$$f'(x) = 18x^2 - 152x + 283$$

$$f'(1.2) = 126.52$$

$$f'(7) = 101$$

$$f''(x) = 36x - 152$$

$$f''(0) = -152$$

Partial derivative

14 May

Given $f(x,y)$

① Partial derivatives for (x):

F.D: $\frac{df(x,y)}{dx} = \frac{f(x+h,y) - f(x,y)}{h}$

B.D: $\frac{df(x,y)}{dx} = \frac{f(x,y) - f(x-h,y)}{h}$

C.D: $\frac{df(x,y)}{dx} = \frac{f(x+h,y) - f(x-h,y)}{2h}$

h: step size

② Partial derivatives for (y)

F.D: $\frac{df(x,y)}{dy} = \frac{f(x,y+h) - f(x,y)}{h}$

B.D: $\frac{df(x,y)}{dy} = \frac{f(x,y) - f(x,y-h)}{h}$

C.D: $\frac{df(x,y)}{dy} = \frac{f(x,y+h) - f(x,y-h)}{2h}$

14 may

Example :

$F(x,y) = x^3 + y^2$, Find the more accurate approximate $dF(x,y)/dx$ & $dF(x,y)/dy$ at point $(1,2)$

$h = 0.1$

Solution:

$$\rightarrow \frac{dF(x,y)}{dx} = \frac{F(1.1,2) - F(0.9,2)}{2(0.1)}$$

$$= \frac{5.331 - 4.729}{0.2} = 3.01 \quad \#$$

$$\rightarrow \frac{dF(x,y)}{dy} = \frac{F(1,2.1) - F(1,1.9)}{2(0.1)}$$

$$= \frac{5.41 - 4.61}{(0.2)} = 4.00 \quad \#$$

14 may

③ Partial derivatives for (x & y)
(mixed derivative)

Centered mixed Derivative:

$$\frac{\partial^2 f(x,y)}{\partial x \partial y} = \frac{f(x+h, y+h) - f(x+h, y-h) + f(x-h, y+h) - f(x-h, y-h)}{4h^2}$$

Example: find $\frac{\partial^2 f(x,y)}{\partial x \partial y} = x^3 + y^3$ at Point (1,2), with a step size of 0.1

Solution:

$x-h = 0.9$
$x+h = 1.1$
$y-h = 1.9$
$y+h = 2.1$

$$\frac{\partial^2 f(x,y)}{\partial x \partial y} = \frac{f(1.1, 2.1) - f(1.1, 2.1) - f(0.9, 2.1) + f(0.9, 1.9)}{4(0.1)^2}$$

= Zero

Differential Equations

19 May

① Ordinary Differential Equations:

$\frac{df}{dx}, \frac{df}{dy}, \frac{df}{dxdy}$

② Partial Differential Equations (PDE):

$\frac{df}{dx}, \frac{df}{dy}, \frac{df}{dxdy}$

① order of D.E
② degree of D.E

$$\rightarrow \frac{d^3f}{dx^3} + \frac{d^2f}{dx^2} + C_1 = C_2$$

order = 3 highest derivative

degree = 1 Power of the highest derivative

Example:

$$\frac{d^4y}{dx^4} + \left(\frac{d^2y}{dx^2}\right)^2 - 3\frac{dy}{dx} + y = 0$$

order = 4
degree = 1

$$\frac{d^2y}{dx^2} + \left(\frac{d^2y}{dx^2}\right)^4 = k^2\left(\frac{d^3y}{dx^3}\right)^2$$

order = 3
degree = 2

19 May

Method of solution :

Runge - Kutta method :

The general form of this method is :

$$y_{i+1} = y_i + \phi(x_i, y_i, h)$$

$$\phi = a_1 k_1 + a_2 k_2 + \dots + a_n k_n$$

$a_1, a_2, a_3, \dots, a_n$ constants

$k_1, k_2, k_3, \dots, k_n$ R.K slopes

$$k_1 = hF(x_i, y_i)$$

$$k_2 = hF(x_i + ph, y_i + q_{11} k_1)$$

$$\vdots$$

k_n

$$a_1 + a_2 = 1.0$$

$$a_2 p_1 = 1/2$$

$$a_2 q_{11} = 1/2$$

19 May

Runge kutta Eqns.

1st order R.k ⟷ Euler method
(توافق وكده)

2nd order R.k , 3rd order R.k , 4th order R.k

1st order R.K method

$y_{i+1} = y_i + f(x_i, y_i)\,h$

h: step size

$y_1 = y_0 + f(x_0, y_0)\,h$

$y_2 = y_1 + f(x_1, y_1)\,h$

Example: solve the differential Eqns.

$$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$$

$f(x, y)$

sinexact = $f'(x, y)$

From $x = 0$ to $x = 4$ with $h = 0.5$ and initial $x_0 = 0$ $y_0 = 1.0$ using R.k

→

19 May

Solution:

exact: $y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x$

$y_0 \leftarrow$

$\rightarrow y_{i+1} = y_i + f(x_i, y_i)h$

$y(0.5) = y(0) + (f(0,1)) * (0.5)$

$\downarrow$
x_0

$y(0.5) = 1.0 + 8.5(0.5) = 5.25$

$\rightarrow$ exact value for $x = 1/2 \rightsquigarrow x_0 + h$

$y = 3.22$

$\hookrightarrow f(1/2)$

$$\varepsilon\% = \left|\frac{3.22 - 5.25}{3.22}\right| * 100\% = 63\%$$

$y_1 = y(0.5) + f(x, y)(0.5) = 5.875$

$\downarrow$ $\downarrow$
0.5 $y(0.5)$

exact value = 3

$$\varepsilon\% = \left|\frac{3 - 5.875}{3}\right| * 100\% = 96\%$$

↓ وأخي تتابع بنفس الطريقة !

$y_{1.5} = y(1) + f(1, 5.875)(0.5) = 5.125$

Lecture 22

21 May

1st order $y_1 = y_0 + \underbrace{hf(x_0, y_0)}_{k_1}$

2nd order $y_1 = y_0 + \frac{1}{2}(k_1 + k_2)$

where $k_1 = hf(x_0, y_0)$

$k_2 = hf(x_0 + h, y_0 + k_1)$

3rd order $y_1 = y_0 + \frac{1}{6}(k_1 + 4k_2 + k_3)$

where $k_1 = hf(x_0, y_0)$

$k_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$

$k_3 = hf(x_0 + h, y_0 + k^*)$

$k^* = hf(x_0 + h, y_0 + k_1)$

4th order $y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

where $k_1 = hf(x_0, y_0)$

$k_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$

$k_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2)$

$k_4 = hf(x_0 + h, y_0 + k_3)$

21 May

Example: For the differential Eqn.

$\frac{dy}{dx} = x^2 + y^2$ of $y(1) = 1.2$, find $y(1.05)$ using R.k.m 4th order.

Solution:

$f(x, y) = x^2 + y^2$

$\rightarrow x_0 = 1$

$y_0 = 1.2$

$h = 1.05 - 1 = 0.05$

$k_1 = 0.05\ (f(1.0, 1.2)) = 0.122$

$k_2 = 0.05\left(f\left(1 + \frac{0.05}{2}, 1.2 + \frac{1}{2}(0.122)\right)\right)$

$k_2 = 1.32$

$k_3 = 0.05\left(f\left(1 + \frac{1}{2}(0.05)\right), 1.2 + \frac{0.132}{2}\right)\right) = 0.1326$

$k_4 = 0.05(f(1 + 0.05, 1.2 + 0.1326)) = 0.1439$

$y(1.05) = 1.2 + \frac{1}{6}(0.122 + 2(0.132) + 2(0.1326) + 0.1439)$

$= 1.3325 \longrightarrow x_0 = 1.05,\ y_0 = 1.3325$

To find $f(1.1)$ ※ $h = 1.1 - 1.05 = 0.05$